

Software Project Management (SPM) (CSC415)

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Chapter 3: Activity Planning and Scheduling

Process and Activity

- A project is composed of a number of interrelated activities.
- A project may start when at least one of its activities is ready to start and will be completed when all of the activities it encompasses have been completed.
- An activity must have a clearly defined start and a clearly defined end-point, normally marked by the production of a tangible deliverable.

Process and Activity

- An activity requires a resource (as most do) then that resource requirement must be forecastable and is assumed to be required at a constant level throughout the duration of the activity.
- The duration of an activity must be forecastable assuming normal circumstances, and the reasonable availability of resources.
- Some activities might require that others are completed before they can begin these are known as precedence requirements.

Introduction to Activity

- Activity, in project management, is defined as the amount of work performed that converts input to appropriate outputs.
- Defining activities refer to the process of identifying as well as documenting actions that need to be implemented and performed in order to produce the deliverables of the project.
- Once the activities have been defined, it's up to the project manager and other stakeholders to sequence them.

Introduction to Activity

- Activities are typically tracked with:
 - ***Network diagram***, which represents all the activities for a project in a sequential, workflow format, or
 - ***Gantt chart***, which represents tasks via horizontal bars that demonstrate their length and duration.

Activity Attributes

- Activity attributes are details of project activities which are used to help project planning and scheduling.
- Activity attributes may be captured and logged either manually via a standard form or template or they may be entered into project and scheduling software.
- After the scope of the project is defined, it is necessary to divide the overall statement of work into the list of activities that are to be completed to finish the project.

Activity Attributes

- An activity list is developed based on the work breakdown structure.
- Each activity has some attributes and activity attributes give the description of each activity and identify other aspects associated with it.
- However, the aspects of each activity may change over the course of the project.
- In the initial phases of the project, an activity is described by a unique code/ID or identifier. With time, some more attributes are added or removed.

Activity Attributes

- These include activity durations, relation of the activity with other activities (successor and predecessor), slack, expected completion time/date, required resources to perform the activity, constraints etc.
- Activity attributes can also be used in schedule development and its related reports.

Activity Sequencing

- Sequence activities is the process of identifying and documenting relationships among the project activities.
- In the project management, the key benefit of this type of process is that it defines the logical sequence of work to obtain the greatest efficiency given all project constraints.
- **How to Sequence Activities in a Project?**
 - Sequencing can be performed by utilizing project management software or by using manual or mechanized procedures.
 - The Sequence Activities process concentrates on converting the project activities from a list to a diagram to act as a first step to publish the schedule baseline.

Activity Planning

- An activity plan should provide a means of evaluating the consequences of not meeting any of the activity target dates.
- It must include the guidance as to how the plan might most effectively be modified to bring the project back to target.

Objectives of Activity Planning

- In addition to providing project and resource schedules, activity planning aims to achieve a number of other objectives which may be summarized as follows.
 - ***Feasibility Assessment:***
 - Is the project possible within required timescales and resource constraints?
 - ***Resource Allocation:***
 - What are the most effective ways of allocating resources to the project and when should they be available?
 - Timescale vs resource availability

Objectives of Activity Planning

- ***Detailed Costing:***

- How much will the project cost and when is that expenditure like to take place?

- ***Motivation:***

- Providing targets and monitoring achievement against targets is an effective way of motivating staff, particularly where they have been involved in setting those targets in the first place.

- ***Co-ordination:***

- When do the staff in different departments need to be available to work on a particular project and when do staff need to be transferred between projects?

When to Start Activity Planning

- Planning is an ongoing process of refinement, each iteration becoming more detailed and more accurate than the last.
- During feasibility study and project start up
 - The main purpose of planning will be to estimate timescales and the risks of not achieving target completion dates or keeping within budget
- As the project proceeds beyond the feasibility study, the emphasis will be placed upon the production of activity plans for ensuring resource availability and cash flow control.

When to Start Activity Planning

- Throughout the project, until the final deliverable has reached the customer, monitoring and re-planning must continue to correct any drift that might prevent meeting time or cost targets.

Different Levels of Plans

- ***Project Schedule:***

- Project scheduling is a mechanism to communicate what tasks to be done, which resources are required and time duration of perform that task.
- A project schedule is a document collecting all the work (*start and end (or due date) of task, schedules of the human resources like vacations and leave*) needed to deliver the project on time.
- Following three questions need to be answered before start:
 - What needs to be done?
 - When will it be done?
 - Who will do it?

Different Levels of Plans

- ***Activity Plan:***

- A plan that describes how each activity will be undertaken
- Some assumptions (criteria) need to make that will be relevant when we start to produce an activity plan.
- Any activity that does not meet these criteria must be redefined.
- Ideal Activity Plan
 - *An activity plan without any constraints*
- Risk consideration for each activity
- Resource consideration for whole project
- Schedule production and publication

Identifying Activity

- Essentially there are three approaches to identifying activities that make up a project.
 - *Activity Based Approach*
 - *Product Based Approach*
 - *Hybrid Approach.*

Activity Based Approach

- The activity-based approach consists of creating a list of all the activities that the project is thought to involve.
- This might involve a brainstorming session involving the whole project team or it might stem from an analysis of similar past projects.
- When listing activities, particularly for a large project, it might be helpful to subdivide the project into the main life-style stages and consider each of these separately.
- Generating a task list is to create a ***Work Breakdown Structure (WBS)***.

Work Breakdown Structure (WBS)

- This involves identifying the main (or high-level) tasks required to complete a project and then breaking each of these down into a set of lower-level tasks.
- Activities are added to a branch in the structure if they directly contribute to the task immediately above if they do not contribute to the parent task, then they should not be added to that branch.
- The tasks at each level in any branch should include everything that is required to complete the task at the higher level if they are not a comprehensive definition of the parent task, then something is missing.
- When preparing a WBS, consideration must be given to the final level of detail.

Work Breakdown Structure (WBS)

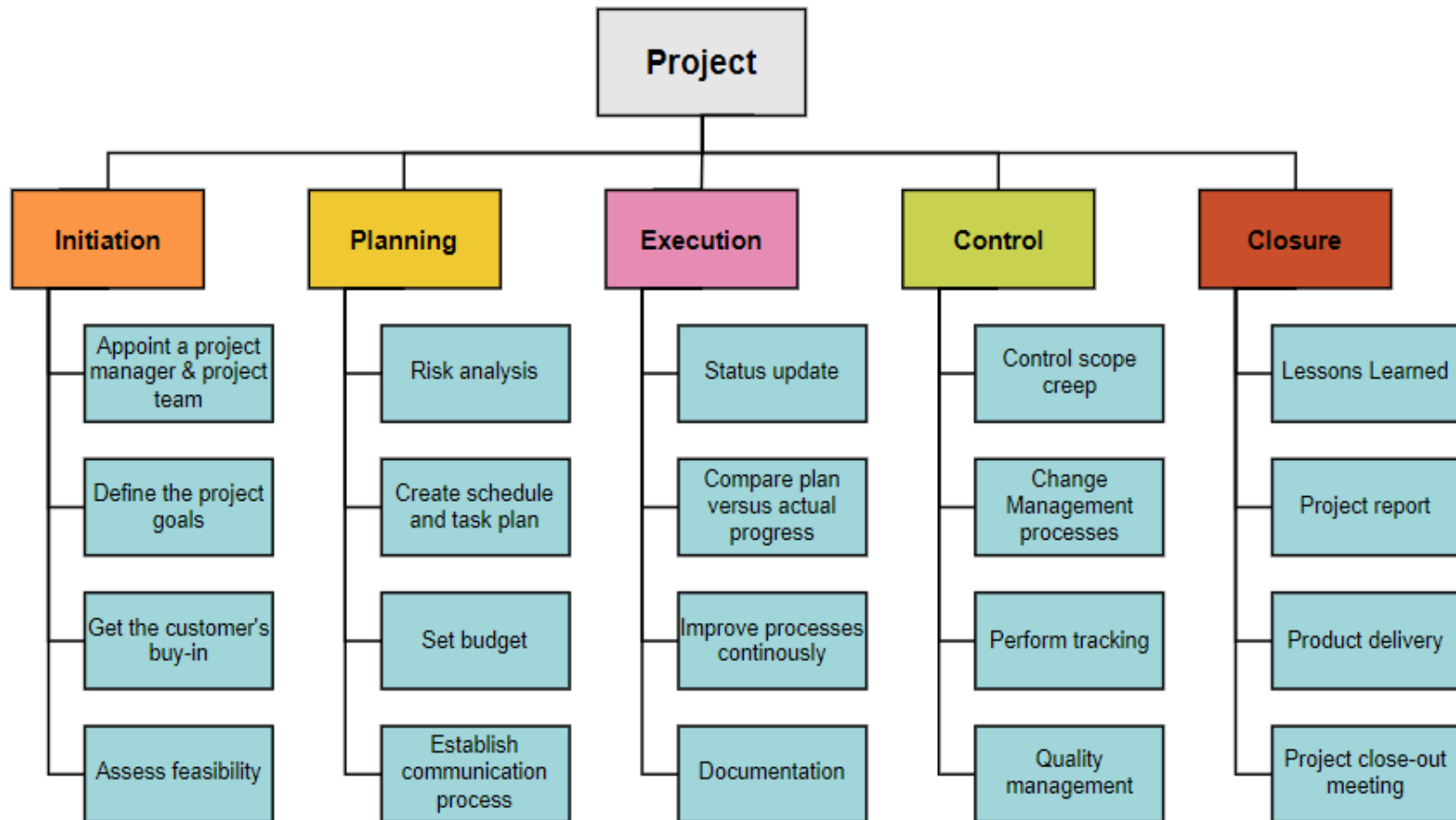


Fig: WBS Structure for Project Management

Work Breakdown Structure (WBS)

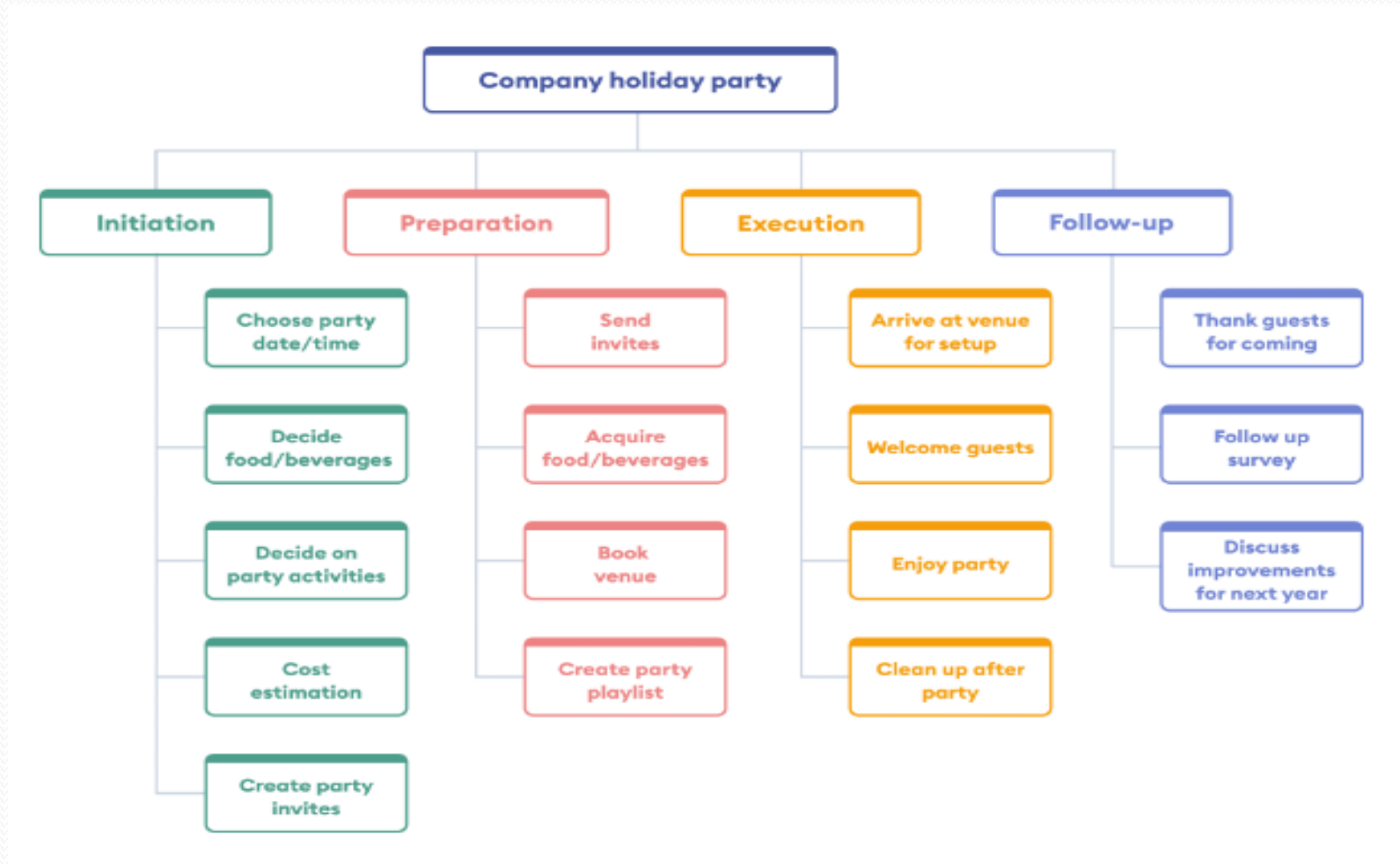


Fig: WBS Structure for Celebration Party

Work Breakdown Structure (WBS)

- ***Advantages:***

- More likely to obtain a task catalogue that is complete and is composed of non-overlapping tasks.
- WBS represents a structure that can be refined as the project proceeds.
- The structure already suggests the dependencies among the activities.

- ***Disadvantage:***

- Very likely to miss some activities if an unstructured activity list is used

Product Based Approach

- It consists of producing a Product Breakdown Structure (PBS) and a Product Flow Diagram (PFD).
- PBS is used to show how a system can be broken down into different products for development
- The PFD indicates, for each product, which other products are required as inputs.
- The PFD can therefore be easily transformed into an ordered list of activities by identifying the transformations that turn products into others.

Product Based Approach

- This approach is particularly appropriate if using a methodology such as Structured Systems Analysis and Design Method (SSADM).
- SSADM clearly specifies, for each step or task, each of the products required and the activities required to produce it.

Product Based Approach

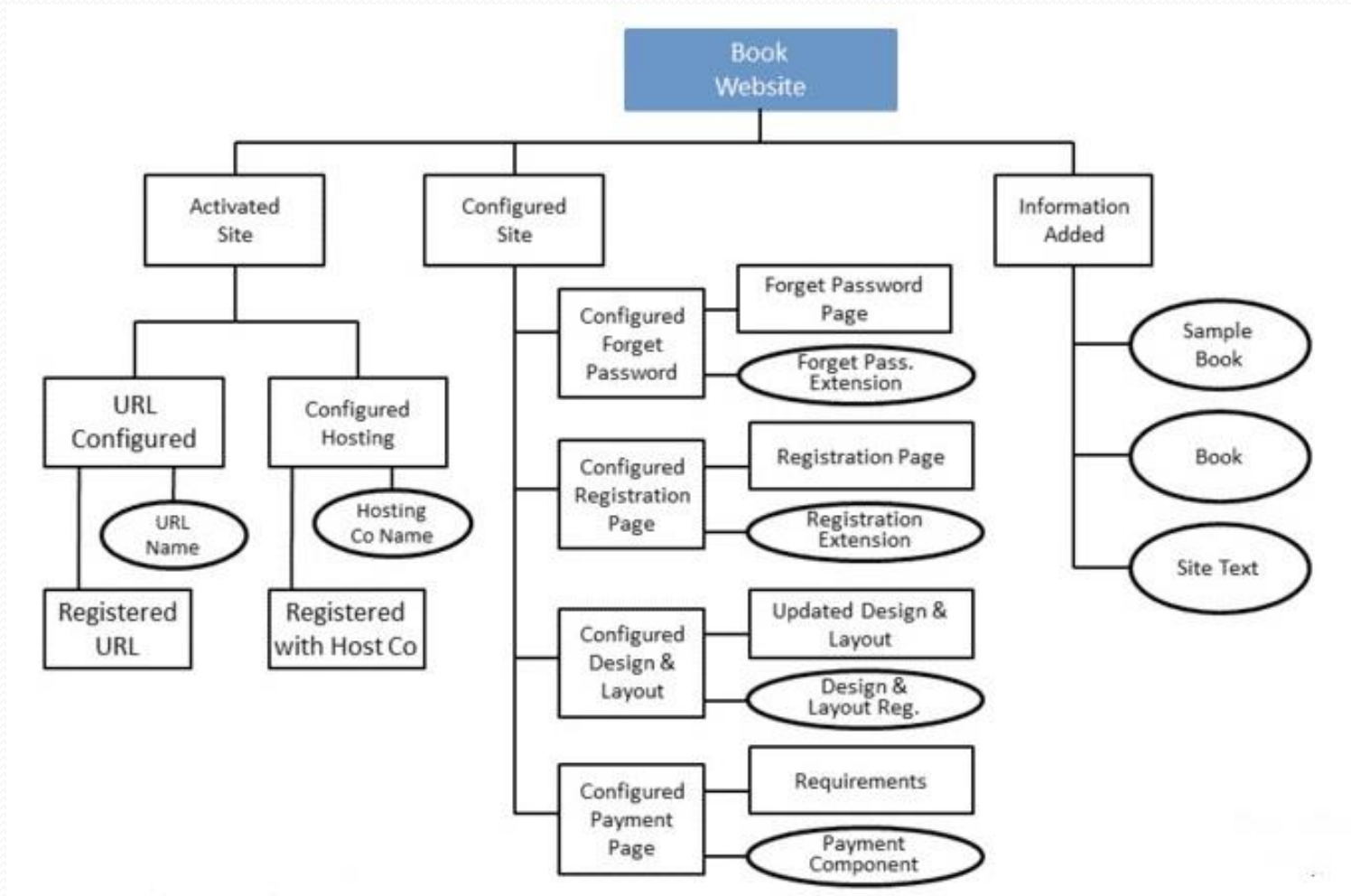


Fig: PBS Structure for Book Website

Product Based Approach

Food Delivery Process



Fig: PFD for Product Delivery System

Hybrid Approach

- Hybrid approach is combination of the activity-based approach and the product-based approach.
- These approach are more commonly used approach than other approaches.
- An alternative WBS based on
 - A simple list of final deliverables.
 - For each deliverable, a set of activities required to produce that product.
- As with a purely activity-based WBS, having identified the activities, we are then left with the task of sequencing them.

Hybrid Approach

- IBM recommend the following five levels should be used in a WBS:
 - ***Level_1: Project.***
 - ***Level_2: Deliverables*** like software, manuals and training courses.
 - ***Level_3: Components*** which are the key work items needed to produce deliverables, such as the modules and tests required to produce the system software.
 - ***Level_4: Work-packages*** which are major work items, or collections of related tasks, required to produce a component.
 - ***Level_5: Tasks*** that will normally be the responsibility of a single person.

Hybrid Approach

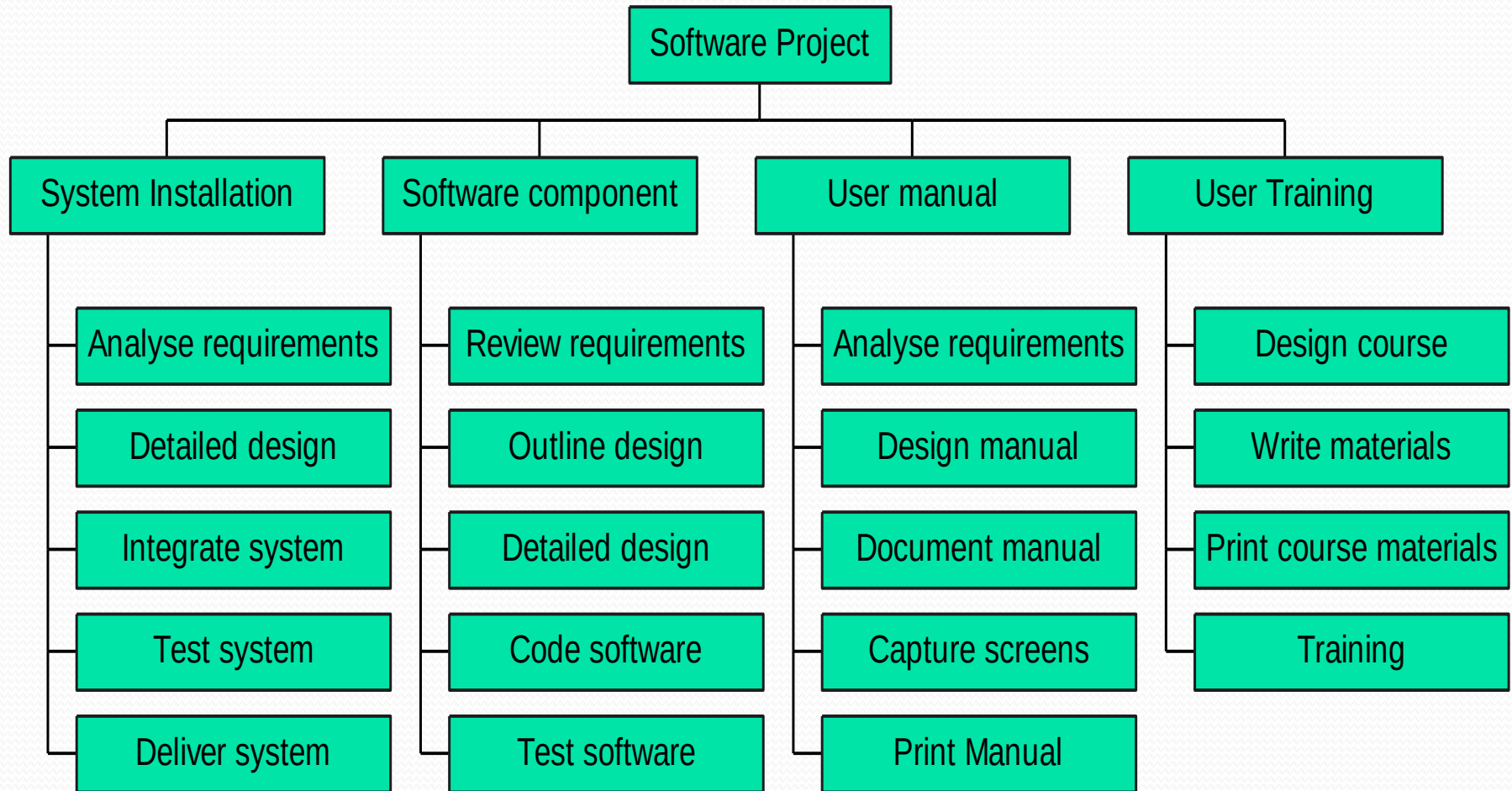


Fig: WBS based on Deliverables

Planning and Scheduling Activities

- Throughout a project, we will require a schedule that clearly indicates when each of the project's activities is planned to occur and what resources it will need.
- Once we have a project plan (or, project schedule), we need to schedule the activities in a project taking into account the resource constraints.
- One way of presenting such a plan is to use a *Bar Chart*.

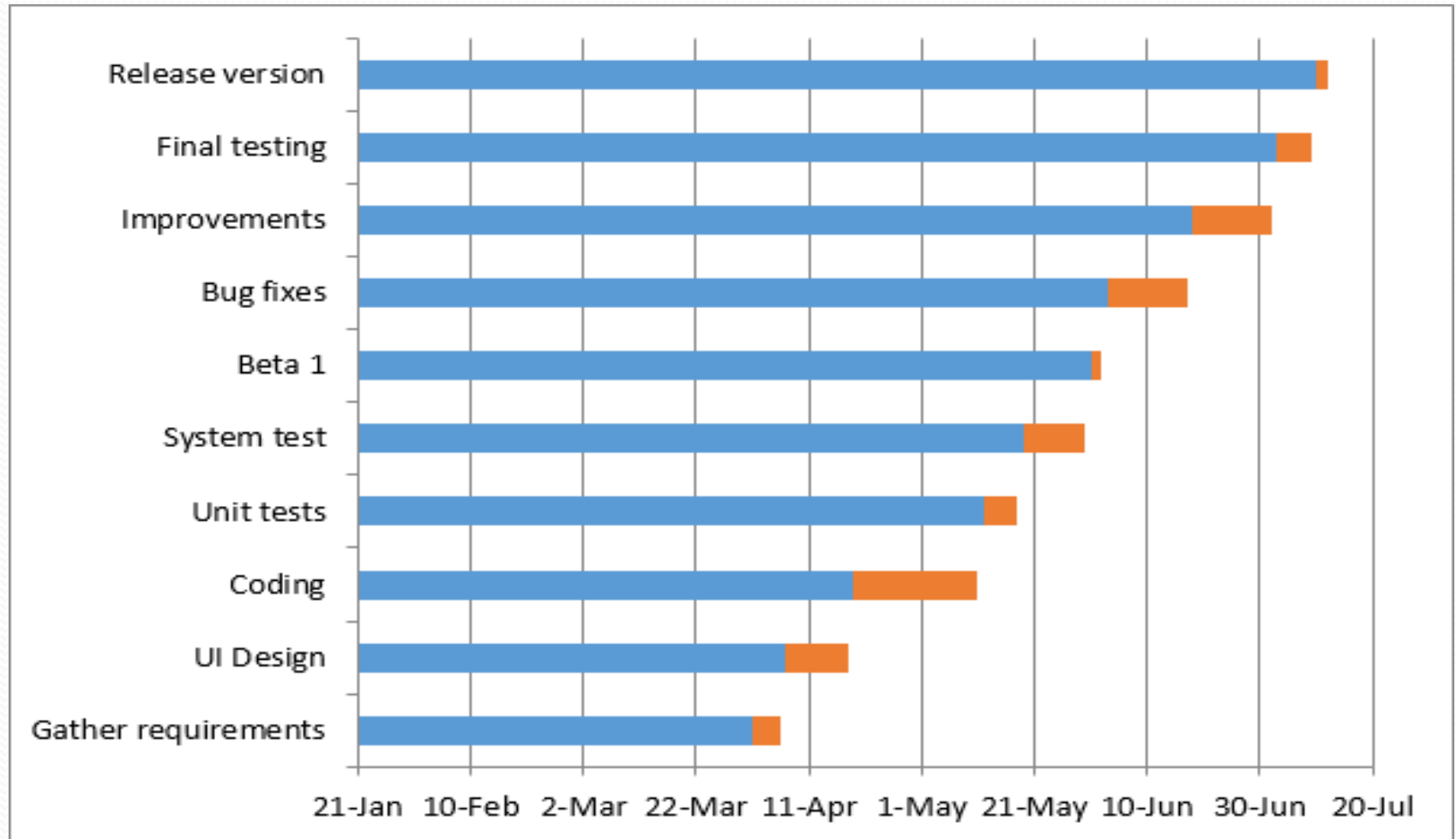
Bar Chart

- A Bar Chart is “a graphic representation of project activities, shown in a time-scaled bar line with no links shown between activities”.
- The bar chart was originally developed by Henry L. Gantt in 1917 and is alternatively called as ***Gantt Chart***.
- It quickly became popular especially in the construction industry because of its ability to graphically represent a project's activities on a time scale.

Bar Chart

- A Gantt chart is used for planning projects of all sizes, and it is a useful way of showing what work is scheduled to be done on a specific day.
- It can also help you view the start and end dates of a project in one simple chart.
- A Gantt Chart reflects the project schedule, which is composed of the WBS, the projected dates of each task are to be started and completed, the milestones, and the resources assigned.

Bar Chart



Bar Chart

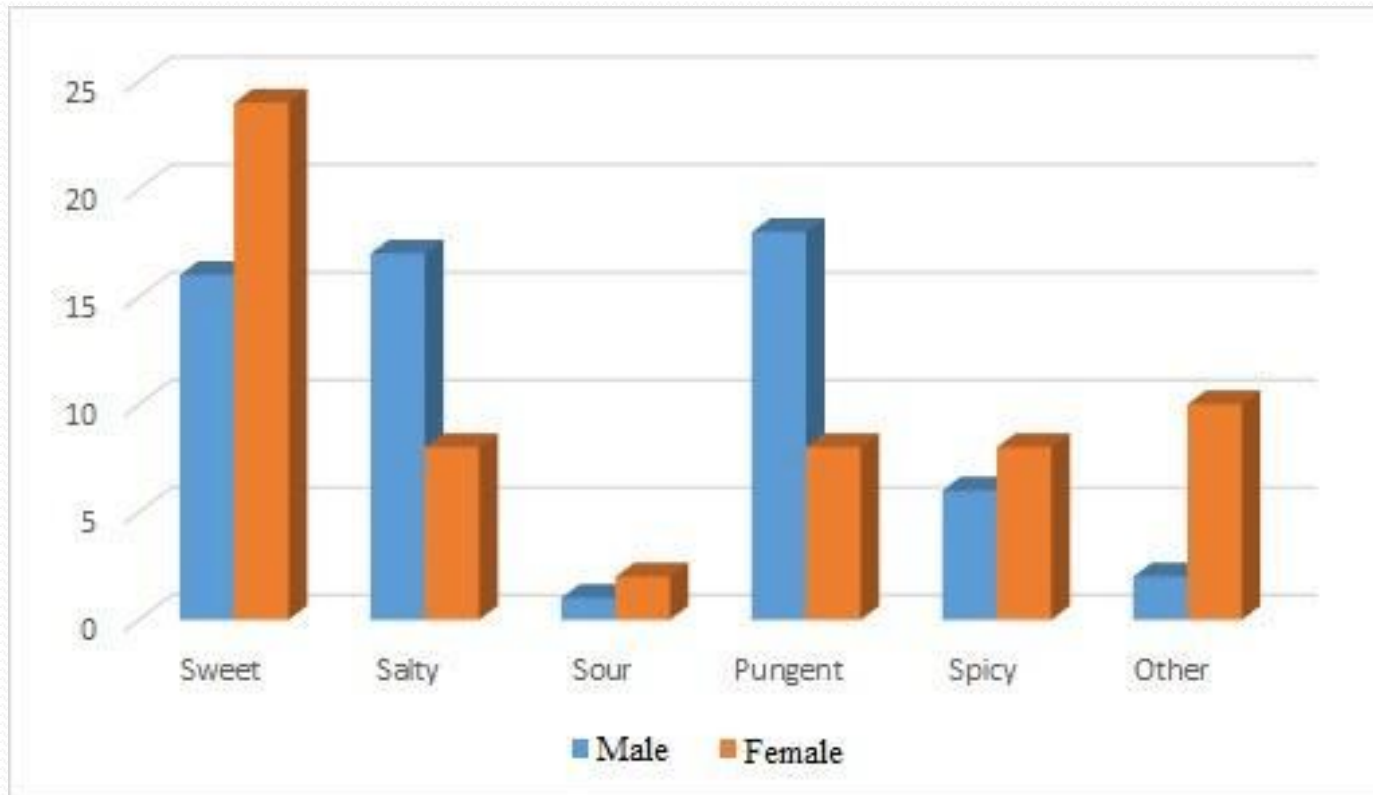


Fig: Bar Graph based on flavor preferences by two genders.

Network Planning Model

- Network Planning Model (NPM) is a critical path model that shows the sequential dependencies among activities in a project.
- It permits the calculation of the earliest project completion date.
- It is a project scheduling model, that models the project activities and their relationships as a network, where the time flows from left to right.

Network Planning Model

- These project scheduling techniques model the project's activities and their relationships as a network.
- Network Planning Model are the method that is adopted in the majority of computer applications currently available.

Network Diagram

- *Network diagrams* are the preferred technique for showing activity sequencing in Network Planning Model.
- A network diagram is a schematic display of the logical relationships among, or sequencing of, project activities.
- Two main formats are:
 - *Arrow Diagramming Method (ADM)*
 - *Precedence Diagramming Method (PDM)*

Network Diagram

- **Arrow Diagramming Method (ADM):**
 - *Also called activity-on-arrow (AOA) network diagram.*
 - *Activities are represented by arrows.*
 - *Nodes or circles are the starting and ending points of activities.*
 - *Can only show finish-to-start dependencies.*
- **Precedence Diagramming Method (PDM):**
 - *Activities are represented by boxes.*
 - *Arrows show relationships between activities.*
 - *More popular than ADM method and used by project management software.*
 - *Better at showing different types of dependencies.*

Network Diagram

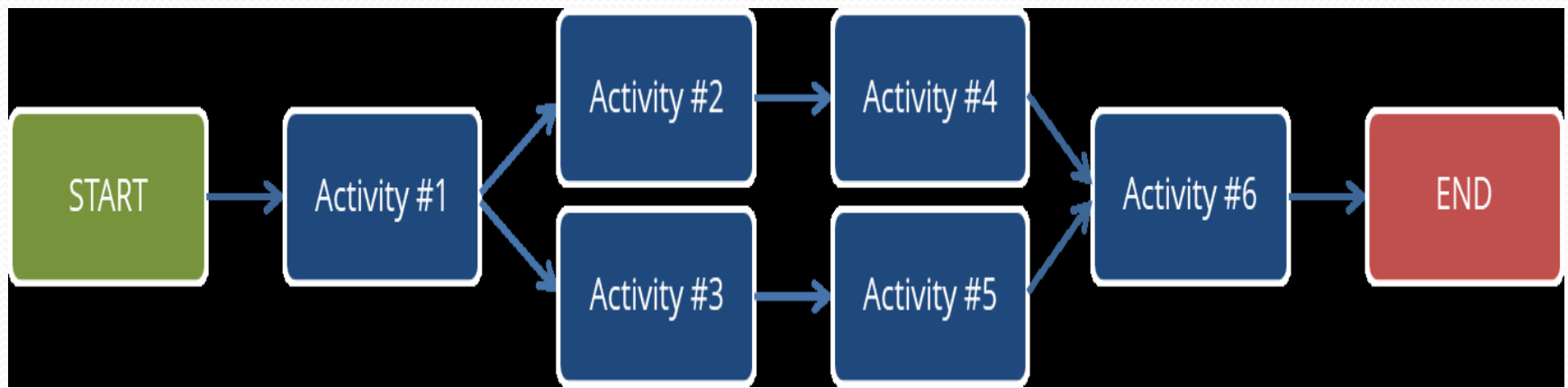
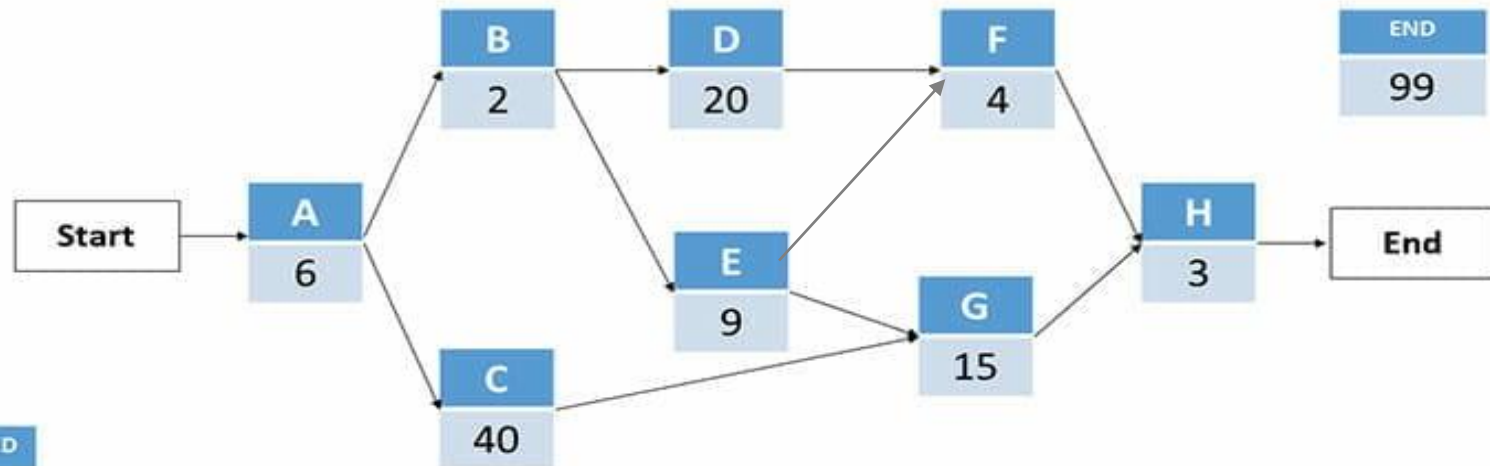


Figure shows a sample network diagram as a result of sequence activities process

Network Diagram

- *Activity #1 must start first.*
- *After Activity #1 finishes, Activity #2 and Activity #3 will begin.*
- *Activity #4 can start only after Activity #2 finishes.*
- *Activity #6 depends on Activity #4 and Activity #5. Therefore, it will start only after these two activities are completed.*
- *After activity #6 completion, the project will end.*

NETWORK DIAGRAM



Activity ID

Duration

Activity	Predecessor	Duration
A		6
B	A	2
C	A	40
D	B	20
E	B	9
F	D,E	4
G	C,E	15
H	F,G	3

Network Planning Model_Techniques

- These techniques are originally developed in 1950's and best known among them are:
 - *Critical Path Method (CPM)*
 - *Program Evaluation Review Technique (PERT)*
 - *Precedence Diagramming Method (PDM)*

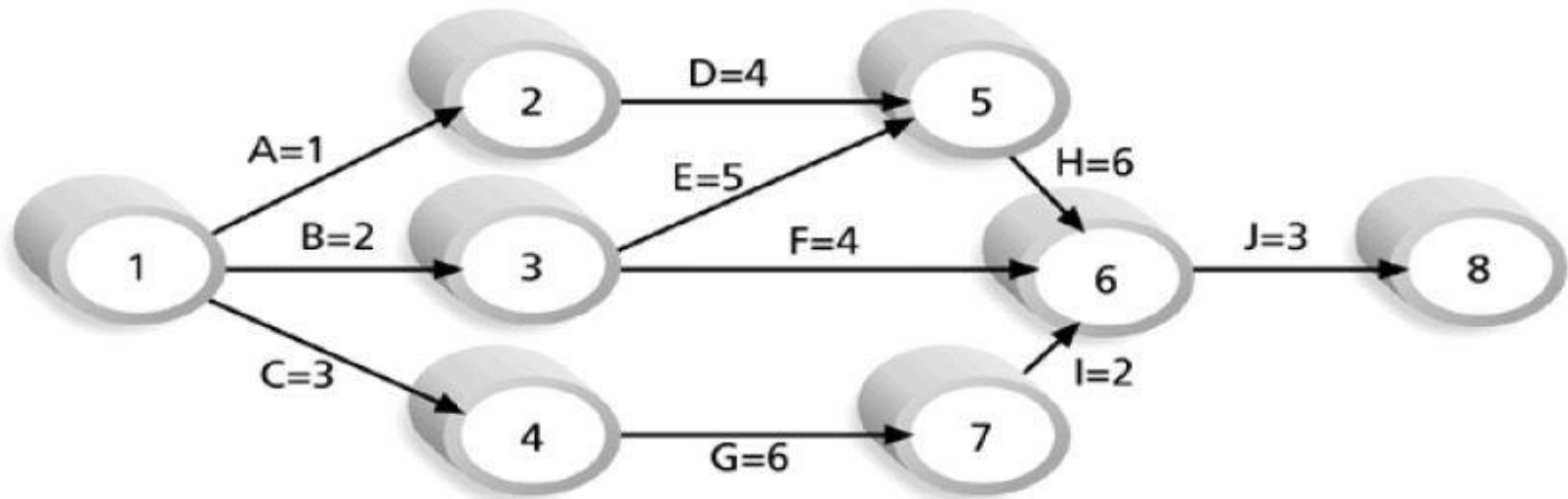
Critical Path Method (CPM)

- CPM is a network diagramming technique used to predict total project duration.
- The critical path is a method used to estimate the minimum project duration and determine the amount of scheduling flexibility on the logical network paths within the schedule model.
- This schedule network analysis technique calculates the *early start*, *early finish*, *late start*, and *late finish* dates for all activities.
- Such calculation is done considering no resource limitations by performing a **forward and backward pass** analysis through the schedule network.

Calculation of Critical Path

- Develop a good network diagram.
- Add the duration estimates for all activities on each path through the network diagram.
- The longest path is the critical path.
- If one or more of the activities on the critical path takes longer than planned, the whole project schedule will slip unless the project manager takes corrective action.

Calculation of Critical Path



Note: Assume all durations are in days.

Path 1: A-D-H-J Length = $1+4+6+3 = 14$ days

Path 2: B-E-H-J Length = $2+5+6+3 = 16$ days

Path 3: B-F-J Length = $2+4+3 = 9$ days

Path 4: C-G-I-J Length = $3+6+2+3 = 14$ days

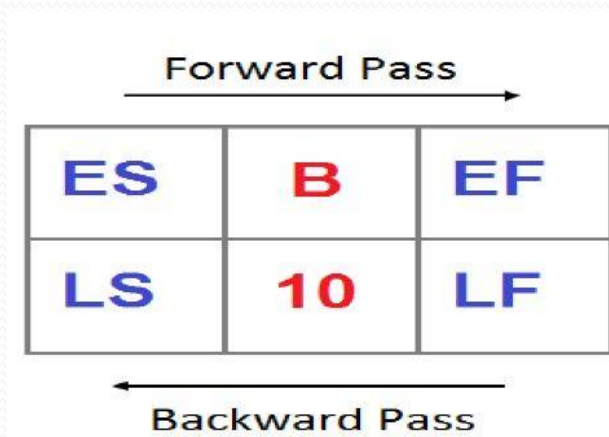
Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

Forward Pass and Backward Pass

- Forward pass is a technique to move forward through network diagram to determining project duration and finding the critical path.
- Whereas backward pass represents moving backward to the end result to calculate late start or to find if there is any slack in the activity.
- To find the critical path we need to plot Early Start (ES), Early Finish (EF), Late Start (LS) and Late Finish (LF) in a network diagram.

Forward Pass and Backward Pass

- Early Start (ES) is plotted on the 1st left corner box at the top. Likewise Early Finish (EF) is plotted on top right corner box.
- Late Finish (LF) is on the right corner box at the bottom and Late Start (LS) is plotted on the left bottom corner box.



Forward Pass and Backward Pass

- Early Start (ES) represents the earliest start of an activity considering the dependency preceding task.
- If an activity is having more than one dependency predecessor, then ES will be the highest Early Finish (EF) of the dependency task.
 - *Early Start = Maximum (or Highest) EF value from immediate Predecessor(s)*
- In order to calculate Early Finish, we use forward pass. Means moving from Early Start towards right to come up with Early Finish of the project.
 - *Early Finish (EF) = ES + Duration*
 - *If Early Start is 6 days and duration is 10 days, $EF = 6 + 10 = 16$ Days*

Forward Pass and Backward Pass

- Late Finish (LF) is the latest date that the activity can finish without causing a delay to the project completion date.
- In order to calculate Late Start (LS), we apply backward Pass moving from Late Finish and deducting from activity duration.
 - ***LS = LF – Duration***
 - ***If Late Finish is 30 days and duration is 10 days then,***

$$LS = 30 - 10 = 20 \text{ Days}$$

Critical Path_Example

- Calculate the critical path for given data using Forward and Backward Pass

Activity	Predecessor	Duration
A	-	5
B	A	7
C	A	4
D	B	10
E	C	3
F	C	5
G	D, E	6
H	F, G	4

Critical Path_Example

Solution,

We first need to draw the network diagram for the activity given considering their predecessor activities.

We have to start from independent activity (i.e. whose precedence is not present).

Here the independent activity is A.

Each activity is then treated as a node of the network and represented as:

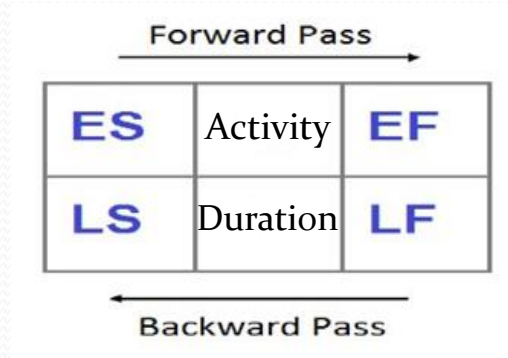


Fig: Activity as Node

Critical Path_Example

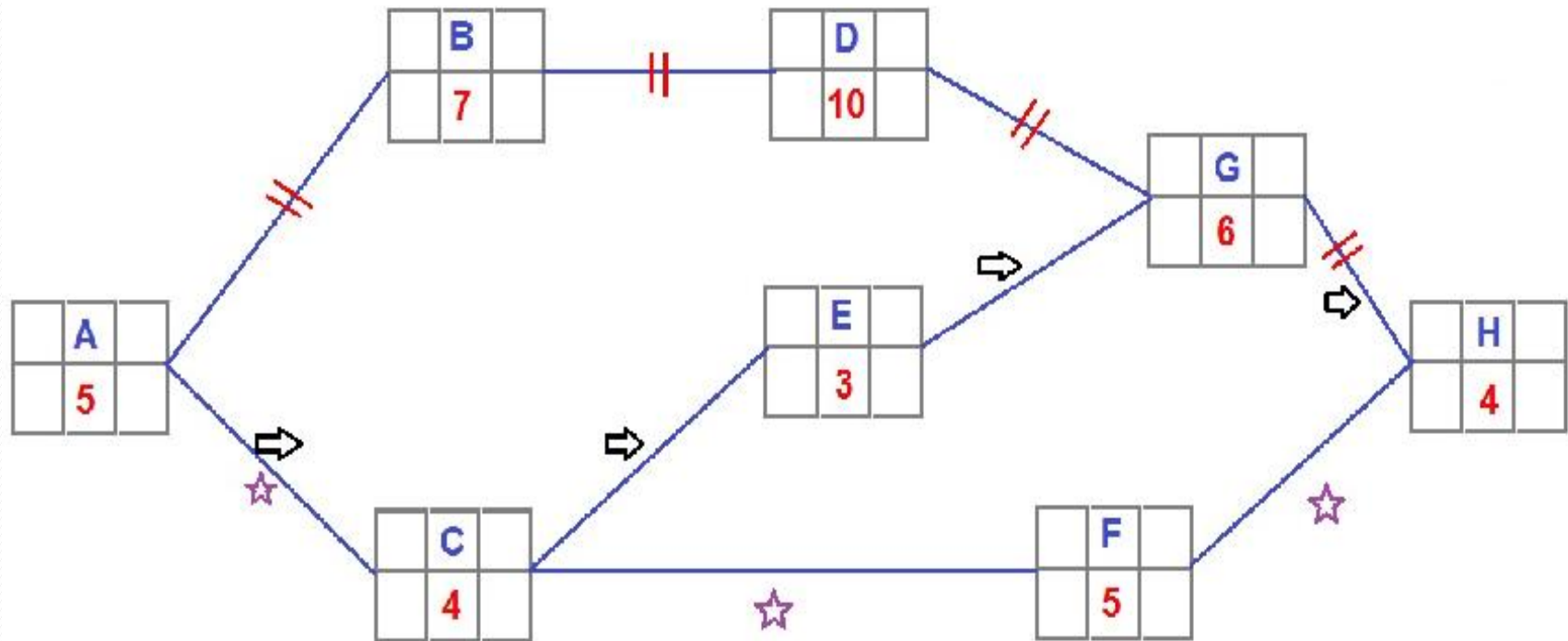
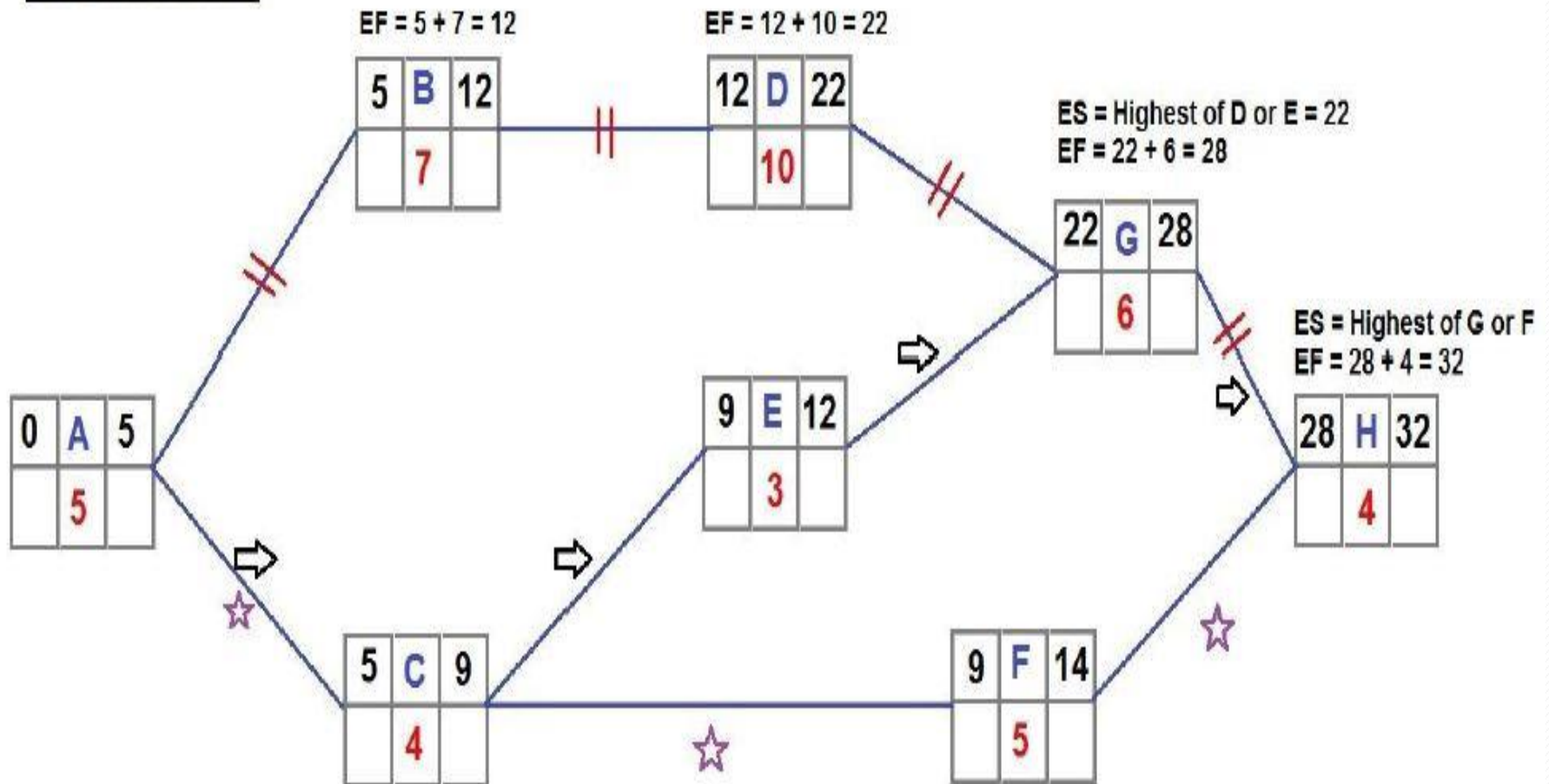


Fig: Network Diagram of Activities

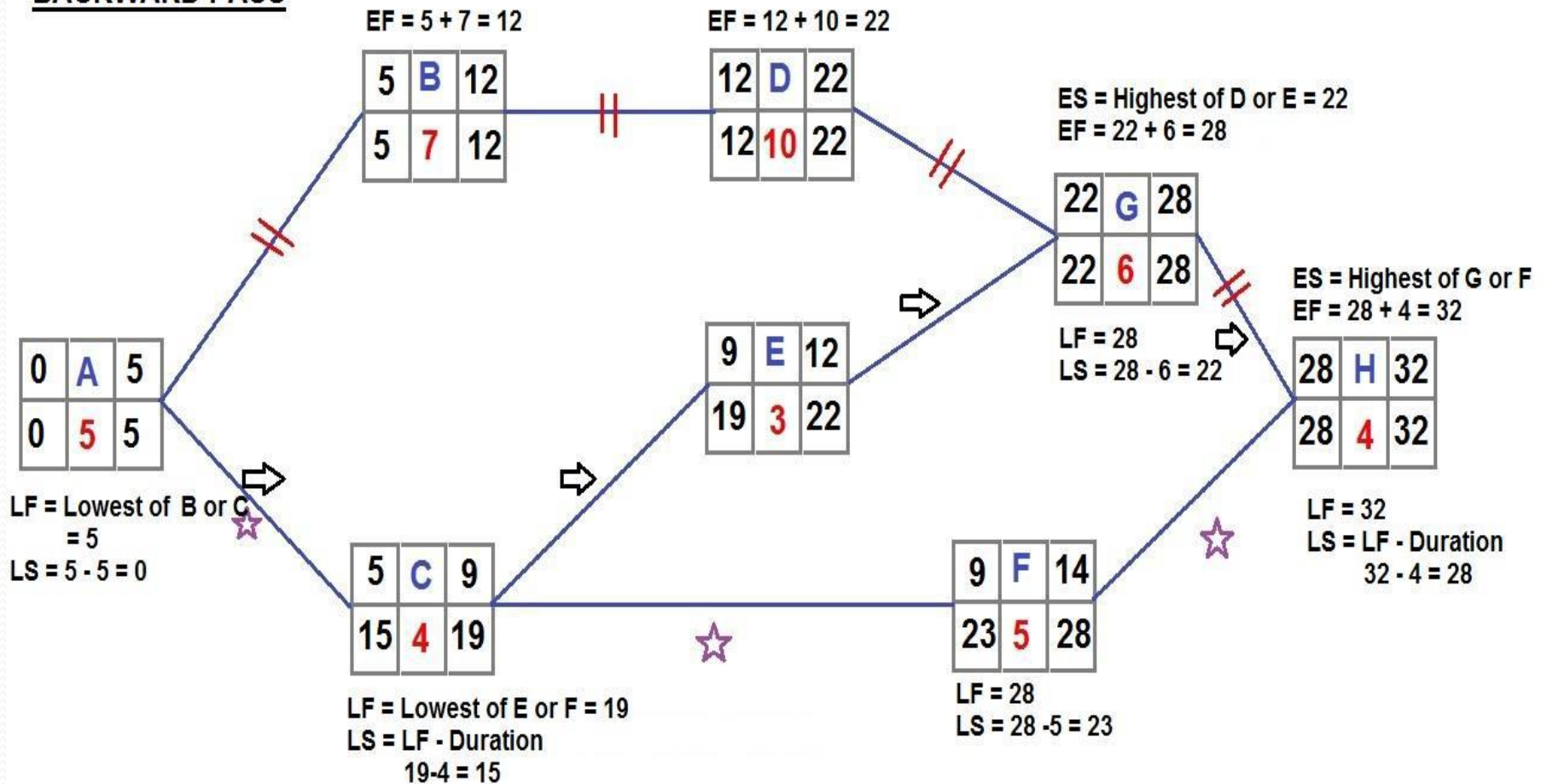
Critical Path_Example

FORWARD PASS



Critical Path_Example

BACKWARD PASS



Critical Path_Example

There are three paths of this project

*To arrive at the **critical path**, add the duration of each activity in the node at its given path.*

The critical path is the one with the longest duration.

$$\mathbf{A + B + D + G + H = 32 \text{ Days}}$$

$$\mathbf{A + C + E + G + H = 22 \text{ Days}}$$

$$\mathbf{A + C + F + H = 18 \text{ Days}}$$

Thus the critical Path is with the longest duration of 32 days i.e.

$$\mathbf{A—B—D—G—H}$$

Critical Path_Question

- Do CPM analysis and identify the critical path for project

Activity	Predecessor	Duration
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4

Program Evaluation Review Technique (PERT)

- Program Evaluation and Review Technique (PERT) is a tool that helps a project manager in project planning and control and enable him in continuously monitoring a project and taking corrective measures wherever necessary.
- In PERT, we assume that it is not possible to have precise time estimate for each activity and instead, probabilistic estimates of time alone are possible.
- A multiple time estimate approach is considered as:
 - Pessimistic Time Estimate (t_p)
 - Optimistic Time Estimate (t_o)
 - Most Likely Time Estimate (t_m)

Program Evaluation Review Technique (PERT)

- The optimistic estimate of time is based on the assumption that an activity will not involve any difficulty during execution and it can be completed within a short period.
- On the other hand, a pessimistic estimate is made on the assumption that there would be unexpected problems during the execution of an activity and hence it would consume more time.
- The most likely time estimate is made in between the optimistic and the pessimistic estimates of time.
- Thus the three estimates of time have the relationship

$$t_o \leq t_m \leq t_p$$

Program Evaluation Review Technique (PERT)

- Now the time estimate (t_e) as the weighted average of these estimates as follows:

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

- PERT is calculated backward from a fixed end date, since deadlines typically cannot be moved.

Program Evaluation Review Technique (PERT)

Example: The following table shows the jobs of a network along with their time estimates.

Activity	Estimated Duration (Weeks)		
	Optimistic (t_o)	Most Likely (t_m)	Pessimistic (t_p)
1-2	1	7	13
1-6	2	5	14
2-3	2	14	26
2-4	2	5	8
3-5	7	10	19
4-5	5	5	17
6-7	5	8	29
5-8	3	3	9
7-8	8	17	32

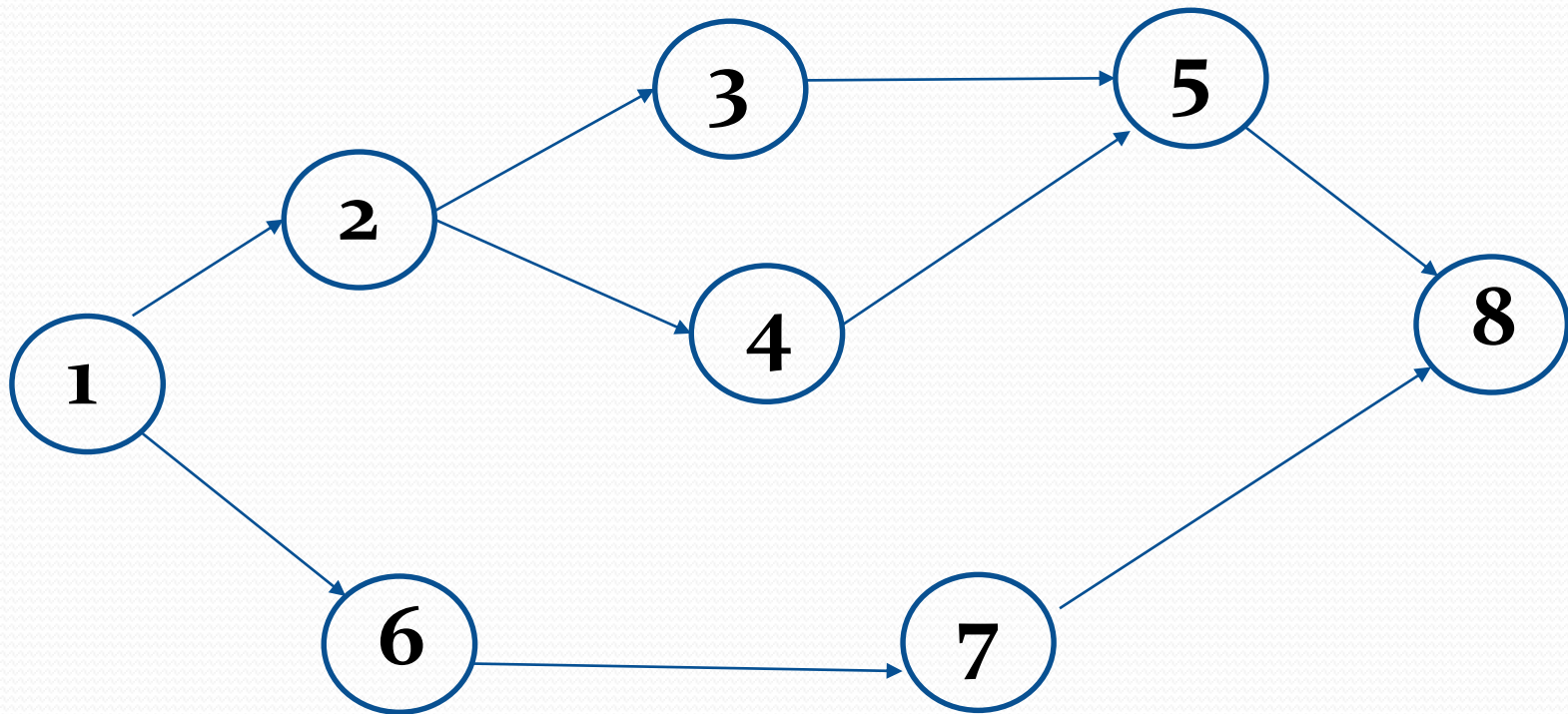
Program Evaluation Review Technique (PERT)

Questions:

- *Draw the project network.*
- *Find the expected duration and variance of each activity.*
- *Calculate the earliest and latest occurrence for each event.*
- *Calculate expected project length.*
- *Calculate the variance and standard deviations of project length.*
- *Find the probability of the project completing in 40 weeks.*

Solution,

1) The network diagram of the job is as follows:

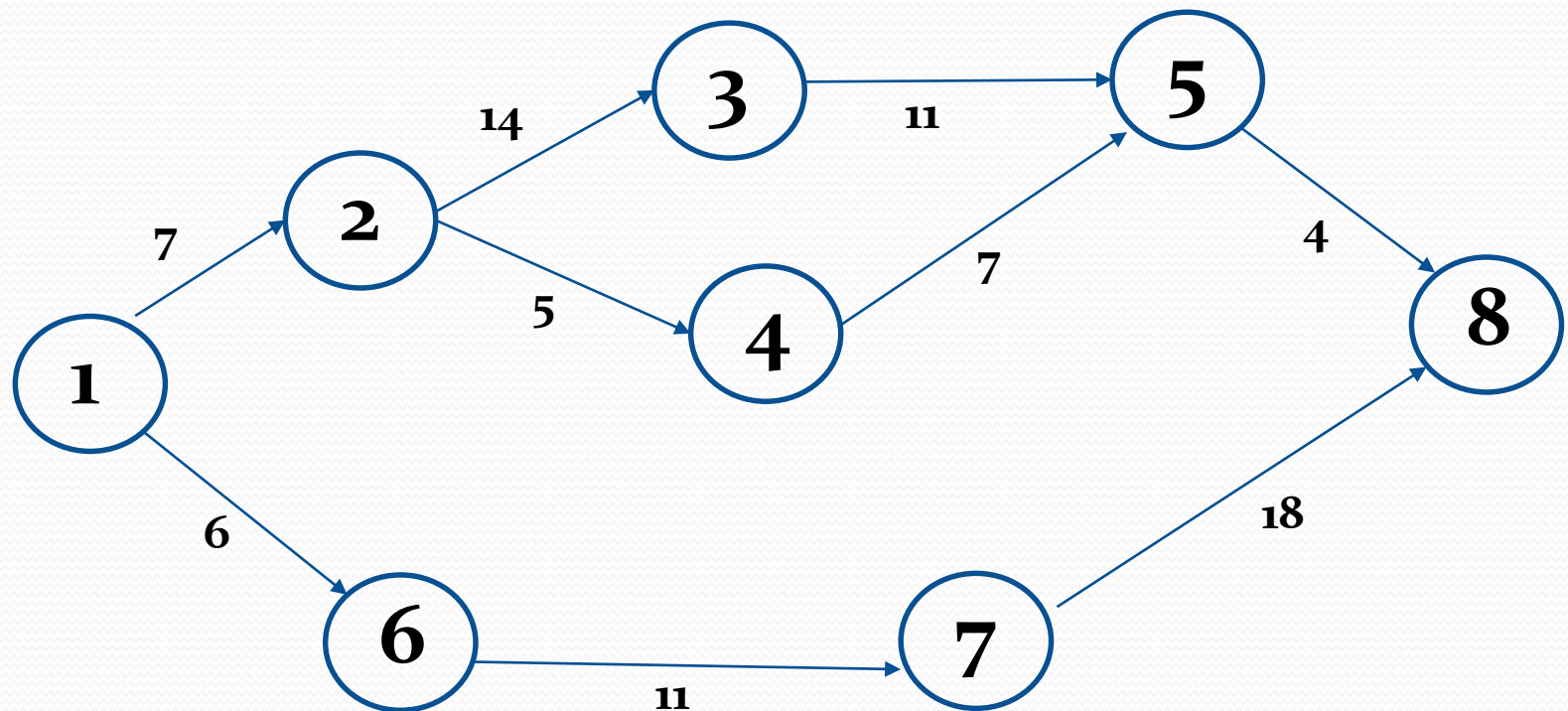


2) Now the expected duration and variance. in the table below:

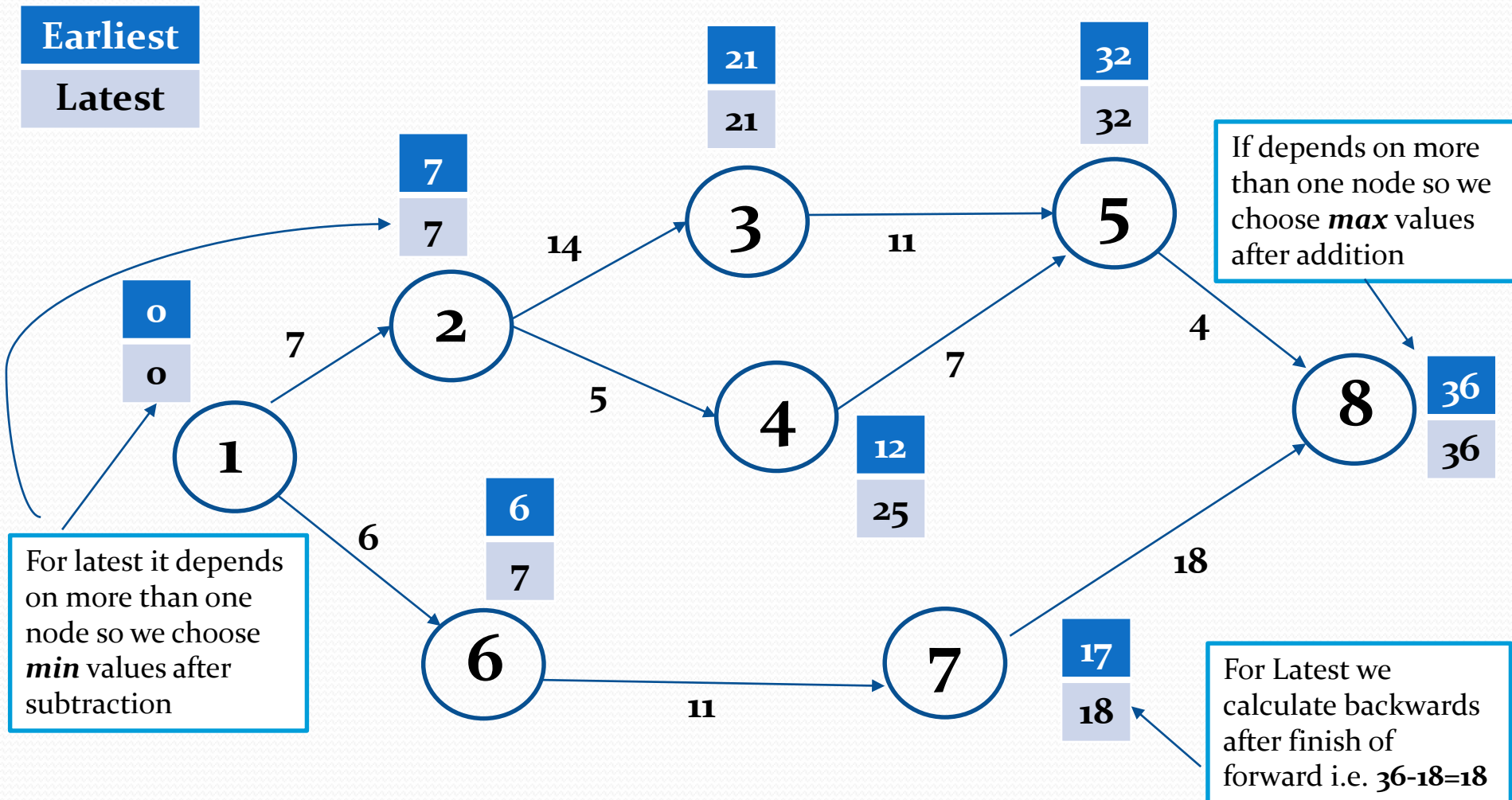
Activity	Estimated Duration (Weeks)			$t_e = \frac{t_o + 4t_m + t_p}{6}$	Variance (σ^2)= $((t_p - t_o)/6)^2$
	Optimistic (t_o)	Most Likely (t_m)	Pessimistic (t_p)		
1-2	1	7	13	7	4
1-6	2	5	14	6	4
2-3	2	14	26	14	16
2-4	2	5	8	5	1
3-5	7	10	19	11	4
4-5	5	5	17	7	4
6-7	5	8	29	11	16
5-8	3	3	9	4	1
7-8	8	17	32	18	16

For Q. No. 3

*We need to draw same network diagram that we draw for Q. No. 1.
Values of t_e are placed to respective paths.
for e.g. activity 1-2 t_e is 7.*



Now we compute earliest and latest occurrence for each event by forward pass and backward pass same as we did for calculation of Early Finish and Late Finish on Critical Path Method.



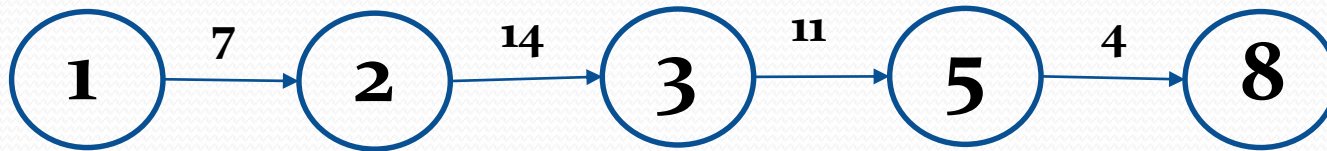
Q.No. 4 We need to find the expected project length

We, have several paths of this project

*To arrive at the **critical path**, we add the duration of each activity in the node at its given path.*

The critical path is the one with the longest duration.

The longest path in this project is:



*So the project length = 7+14+11+4
= 36 weeks*

Q.No. 5 To calculate the variance and standard deviations of project length

We need to add the variance of the critical path.

Since $1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 8$ is the critical path so $1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 5$, $5 \rightarrow 8$ are the activities.

*So, add the values **4, 16, 4, 1** since these are the respective variance of the activities in the critical path.*

$$\begin{aligned}\text{Variance of project length} &= \sum \sigma^2 \text{ for critical path} \\ &= 4 + 16 + 4 + 1 \\ &= 25\end{aligned}$$

Q.No. 6 To Find the probability of the project completing in 40 weeks.

The probability that the project will be completed in 40 weeks is given by $P(z \leq D)$, where $D = \frac{T_s - T_e}{\sigma}$

T_s is the schedule time given in question i.e. 40 weeks.

T_e is the project length calculated earlier i.e. 36 weeks.

$$D = \frac{40 - 36}{5} = 0.8$$

Now, $P(z \leq 0.8) = 0.78814$
 $= 78.814\%$

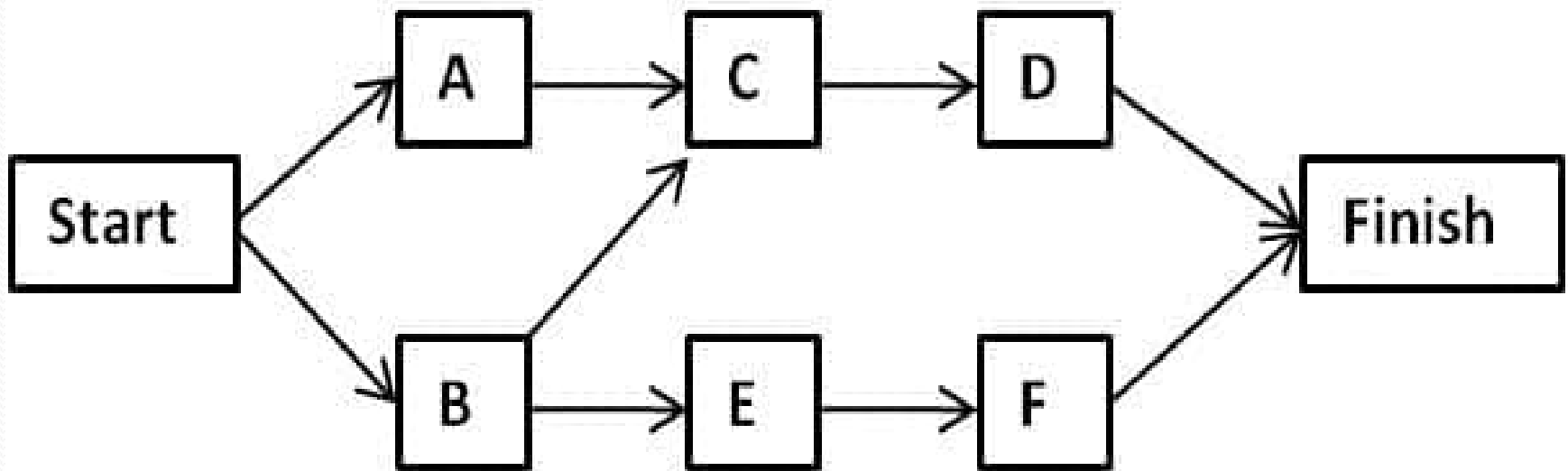
Value from the probability distribution table for 0.8 in .00 i.e. first column

Hence, the probability that the project will be completed in 40 weeks is 78.814%

Precedence Diagramming Method (PDM)

- It is a visual representation technique that represents the activities involved in a project.
- It is a method of constructing a project schedule network diagram that uses nodes to represent activities and connects them with arrows that show the dependencies.
- As, CPM and PERT are limited to "start-end" relationship (i.e. activity B cannot start until activity A is completed).
- PDM was developed subsequent to CPM and PERT techniques and to permit a more accurate representation of relationship among various activities.

Precedence Diagramming Method (PDM)



Precedence Diagram Method (PDM)

Precedence Diagramming Method (PDM)

- The model can reflect 4 types of dependencies
 - ***Finish-to-Start (FS) dependency***
 - If two activities are linked by a finish-to-start dependency, this means that the predecessor activity must be completed before the successor activity can start.
 - ***Finish-to-Finish (FF) dependency***
 - Under this type of logical relationship, a successor activity requires the predecessor activity to be finished before it can be completed.
 - This type of dependency can also coincide with start-to-start relationships of the same activities.

Precedence Diagramming Method (PDM)

- ***Start-to-Start (SS) dependency***
 - If two activities are connected through a start-to-start dependency, this means that the predecessor activity must have started before the successor activity can start.
- ***Start-to-Finish (SF) dependency***
 - This logical relationship requires that a predecessor activity must have started before the successor activity can be finished.
 - In practice, this type of dependency does not occur very often which is also acknowledged by the PMBOK.

Precedence Diagramming Method (PDM)

- ***Benefits***

- *Highlights relations and dependencies among activities.*
- *Identify possible missing activities.*
- *Helps develop overall project schedule.*
- *Good communication tool for project team members*

Precedence Diagramming Method (PDM)

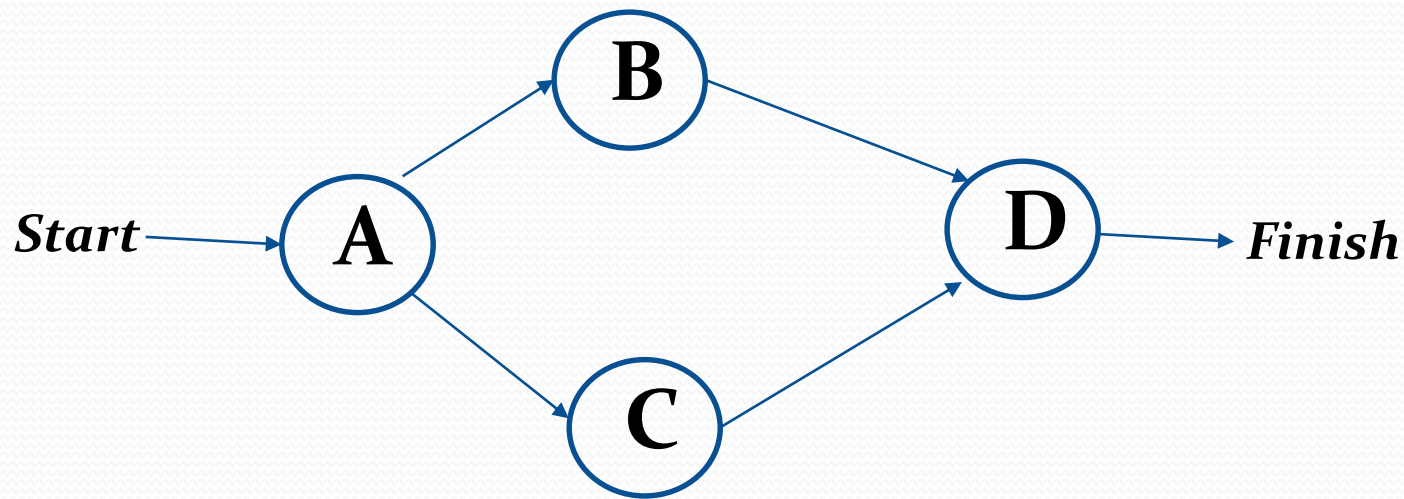
- *Draw, precedence network diagram for the following and identity the critical Path.*

Activity	Precedence	Duration
A	-	5
B	A	3
C	A	5
D	B, C	10

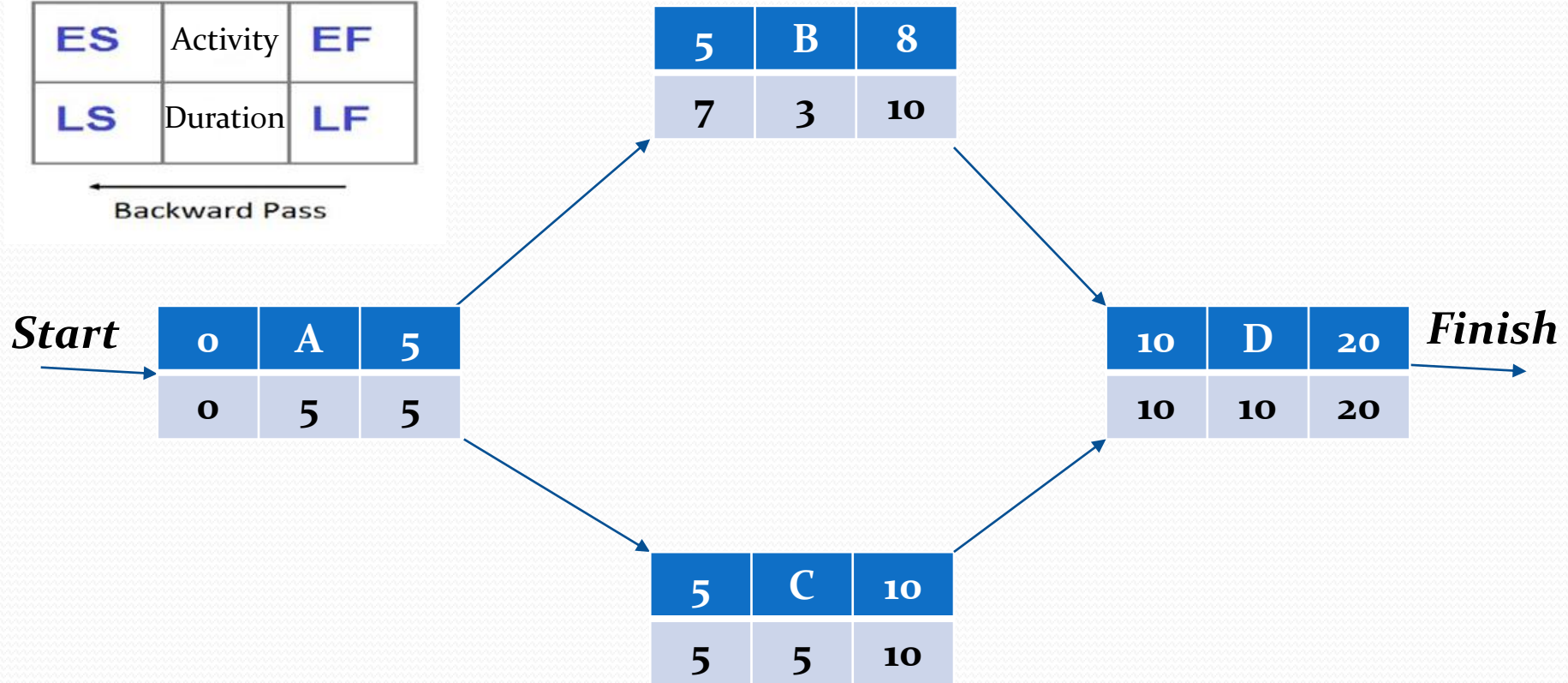
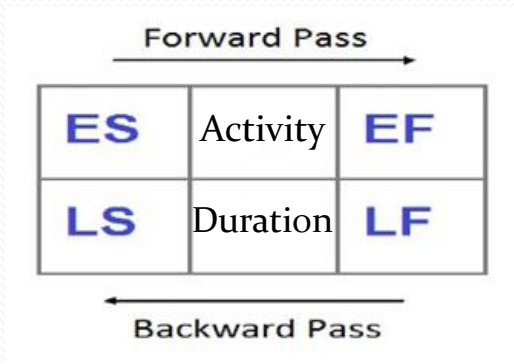
Precedence Diagramming Method (PDM)

Solution,

The network diagram is:



Now, we can do as same as we do in CPM.



Now, the Possible path are; $A \rightarrow B \rightarrow D \rightarrow \text{Finish}$ and $A \rightarrow C \rightarrow D \rightarrow \text{Finish}$
 Here, we will choose the Path $A \rightarrow C \rightarrow D$ because the value of Early Start and Late Start are same or **it is a critical path**. So, we choose that path.

Question

The table below is an example of project specification with estimated activity duration and precedence requirements

Activity	Activity Name	Duration (Weeks)	Precedence
A	Hardware Selection	6	-
B	System Configuration	4	-
C	Install Hardware	3	A
D	Data Migration	4	B
E	Draft Office Procedure	3	B
F	Recruit Staff	10	-
G	User Training	3	E, F
H	Install and Test System	2	C, D

Find the critical path of the project and calculate the earliest completion time of the project.

Shortening Project Duration

- If we wish to shorten the overall duration of a project we would normally consider attempting to reduce activity durations.
- In many cases this can be done by applying more resources to the task - working overtime or procuring additional staff, for example.
 - *Example: The critical path indicates where we must look to save time if we are trying to bring forward the end date of project.*
 - *As we reduce activity times along the critical path, we must continuously check for any new critical path emerging and redirect attention, where necessary.*

Shortening Project Duration

- Parkinson's Law states, “*Work expands to meet the time available for its completion.*”
- The project corollary is that tasks are not completed before their planned finish date.
- Some approaches are:
 - *Establish Scope Before the Project Starts*
 - *Eliminate False Dependencies*
 - *Develop a Comprehensive Testing Strategy*
 - *Save Days*

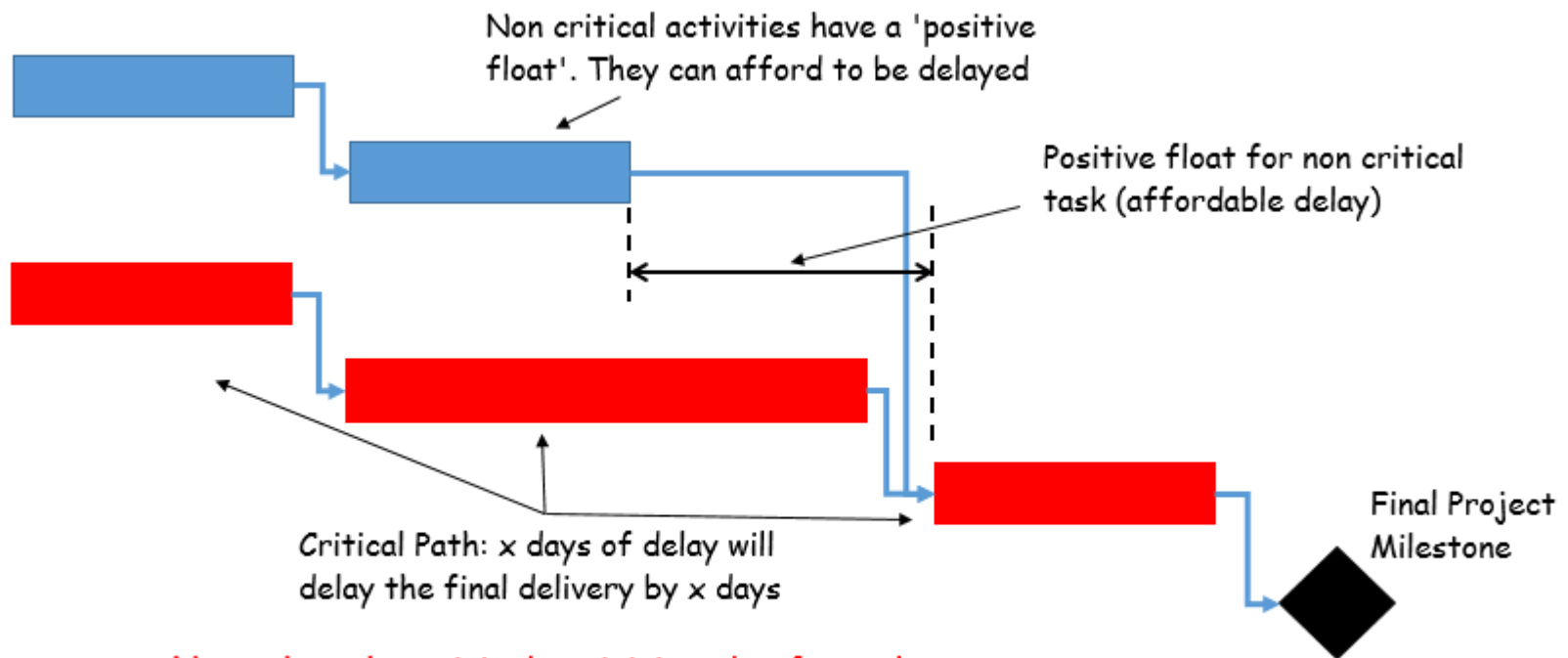
Identifying Critical Activities

- Critical activities are those, for which the start and end dates are strictly defined.
- They are critical in sense that their delay results in delay of whole project.
- The critical path indicates or identifies those activities which are critical to an end date of project, however the activities that are not on critical path may become critical.
- As the project proceeds, activities will require a periodic calculation of network.

Identifying Critical Activities

- As soon as the activities along critical path use up their total float then that path will become a critical path and a number of previous non-critical activities will suddenly become critical.
- Some approaches for identifying critical activities are:
 - *Define high level project roadmap*
 - *Create a detailed work breakdown structure*
 - *Define duration and dependency relationships for project activities*
 - *Identify key milestones*
 - *Use critical path method (CPM) to create a project schedule*
 - *Identify critical activities as per the project schedule*

Identifying Critical Activities



Note that the critical activities also form the longest chain in terms of duration on Gantt chart

Assignment-II

- Differentiate between CPM and PERT.
- Why planning is necessary. Highlight on the steps of activity planning.
- Differentiate between Work break down structure and product breakdown structure with an example.
- Why software project management is a challenging activity? Differentiate between activity-based approach and hybrid-based approach for identifying activities.
- Explain the objectives of activity planning. Explain the Gantt Charts in Scheduling with example.

Submission Deadline: 13th February_2026



End of Chapter 3

Thank You !!!!